

Atelier 5 : Notion d'application Microservices (exemple Books / Customers / Orders)

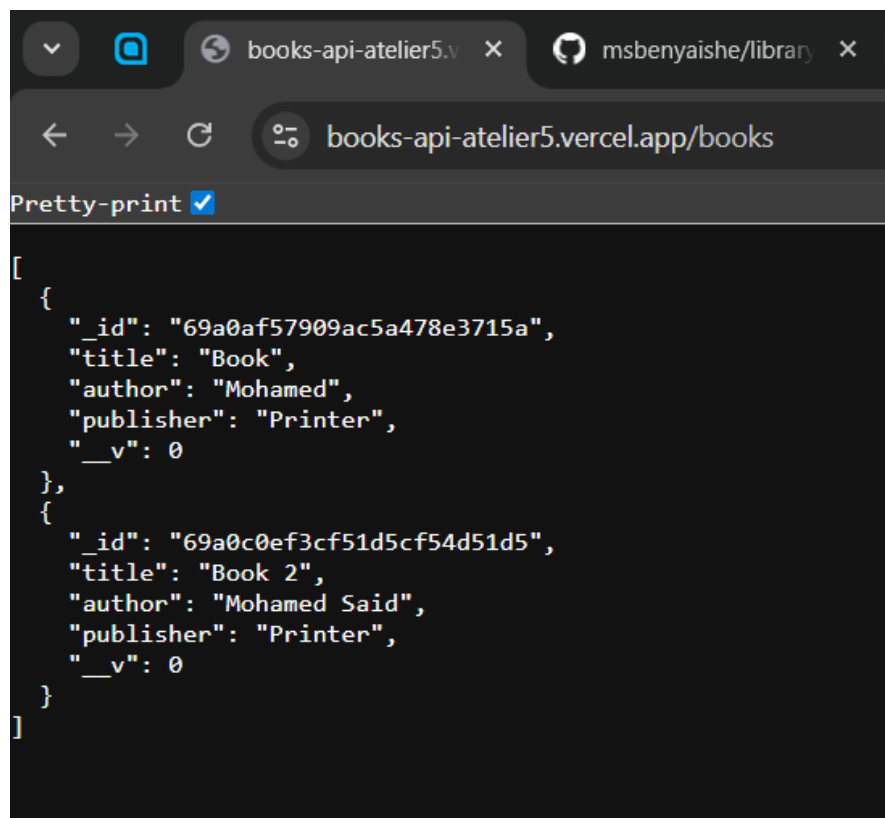
1. Architecture des Microservices

A. Books Service (Service des Livres)

- Rôle : Gère l'inventaire des livres (titre, auteur, ISBN, etc.).
- Technologie : Node.js, Express, MongoDB (Atlas).
- Points clés : Endpoints pour ajouter, lister et supprimer des livres.
- Base de données : Collection books.

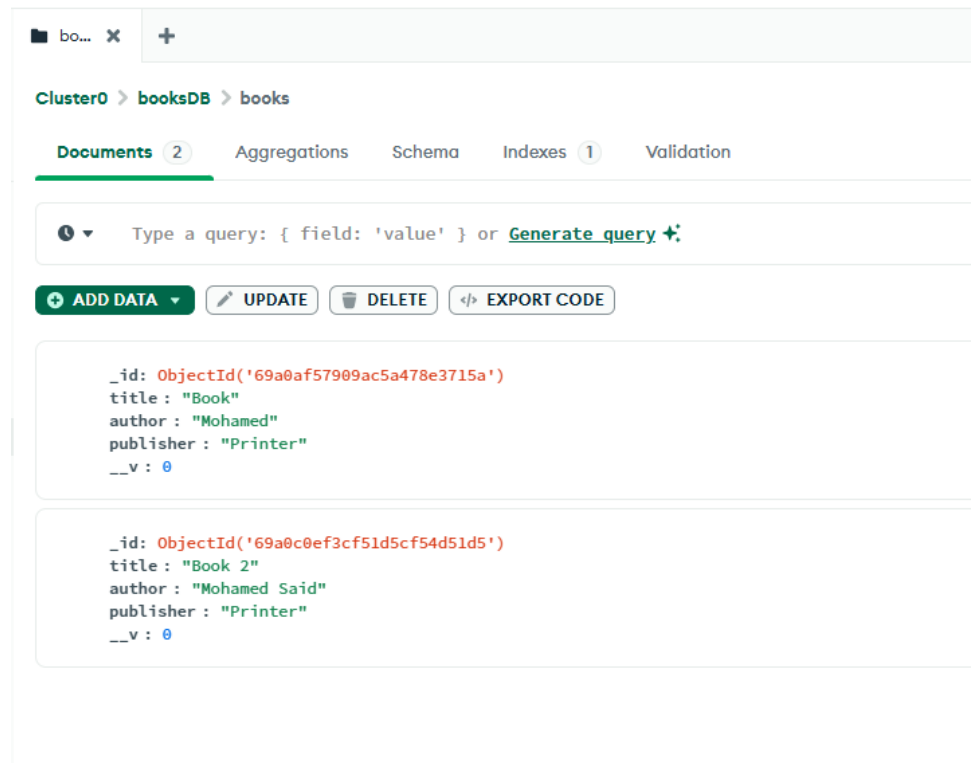
Lien API :

<https://books-api-atelier5.vercel.app/books>



```
[
  {
    "_id": "69a0af57909ac5a478e3715a",
    "title": "Book",
    "author": "Mohamed",
    "publisher": "Printer",
    "__v": 0
  },
  {
    "_id": "69a0c0ef3cf51d5cf54d51d5",
    "title": "Book 2",
    "author": "Mohamed Said",
    "publisher": "Printer",
    "__v": 0
  }
]
```

Collection books :

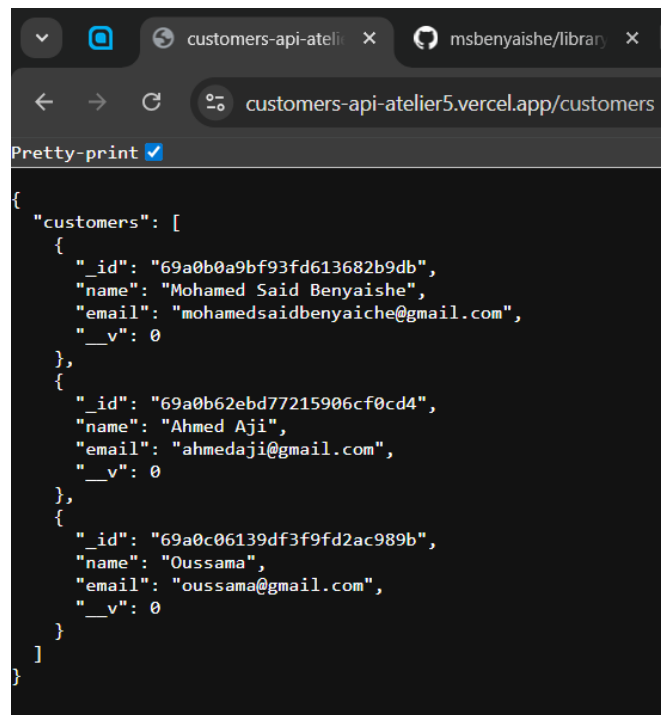


B. Customers Service (Service des Membres)

- Rôle : Gère les informations des membres de la bibliothèque.
- Technologie : Node.js, Express, MongoDB.
- Points clés : Gestion des profils clients (Nom, Email, etc.).
- Base de données : Collection customers.

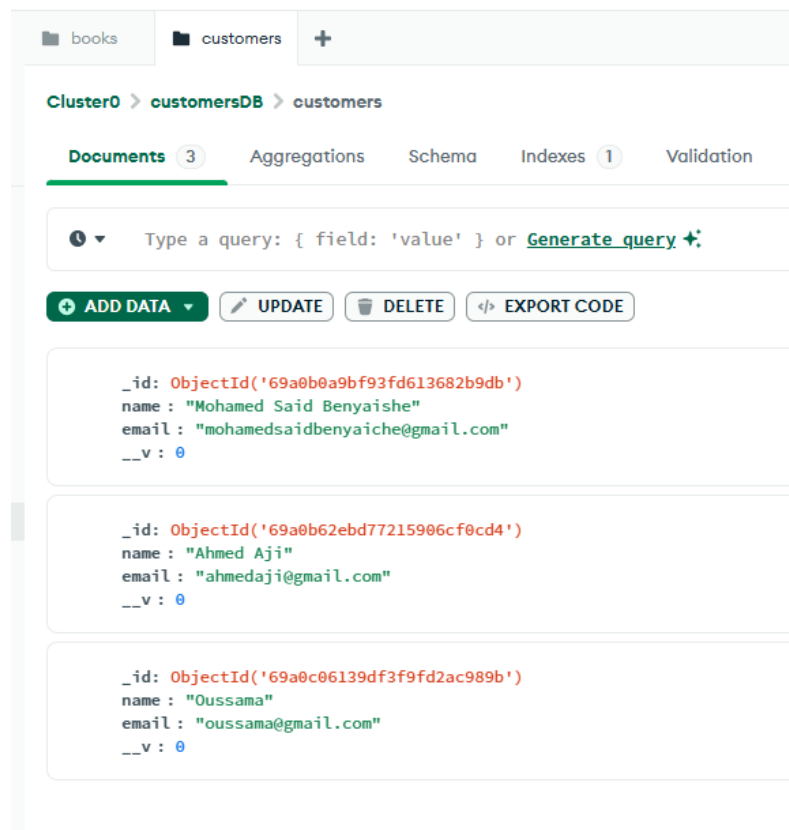
Lien API :

<https://customers-api-atelier5.vercel.app/customers>



```
{
  "customers": [
    {
      "_id": "69a0b0a9bf93fd613682b9db",
      "name": "Mohamed Said Benyaishe",
      "email": "mohamedsaidbenyaiche@gmail.com",
      "__v": 0
    },
    {
      "_id": "69a0b62ebd77215906cf0cd4",
      "name": "Ahmed Aji",
      "email": "ahmedaji@gmail.com",
      "__v": 0
    },
    {
      "_id": "69a0c06139df3f9fd2ac989b",
      "name": "Oussama",
      "email": "oussama@gmail.com",
      "__v": 0
    }
  ]
}
```

Collection customers :



Cluster0 > customersDB > customers

Documents (3) Aggregations Schema Indexes (1) Validation

Type a query: { field: 'value' } or [Generate query](#)

[+ ADD DATA](#) [UPDATE](#) [DELETE](#) [EXPORT CODE](#)

```
_id: ObjectId('69a0b0a9bf93fd613682b9db')
name: "Mohamed Said Benyaishe"
email: "mohamedsaidbenyaiche@gmail.com"
__v: 0
```

```
_id: ObjectId('69a0b62ebd77215906cf0cd4')
name: "Ahmed Aji"
email: "ahmedaji@gmail.com"
__v: 0
```

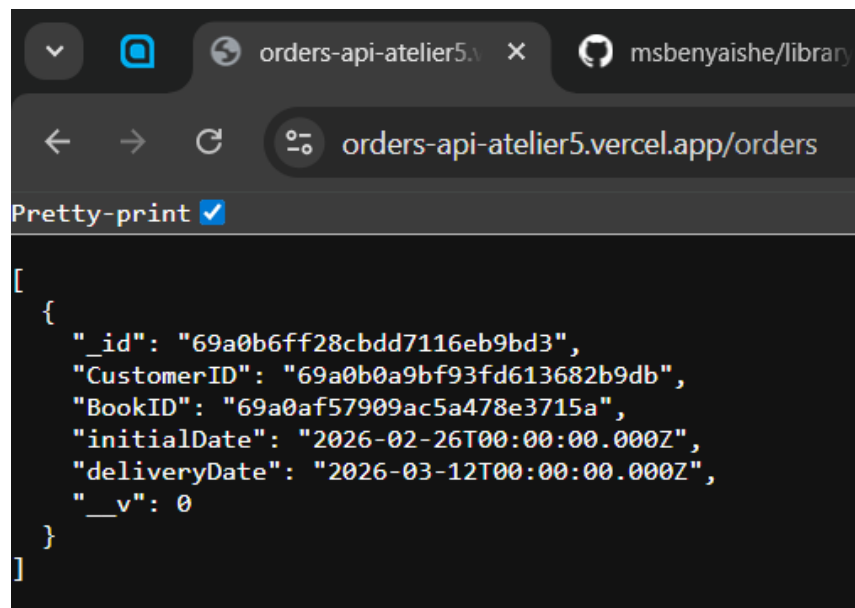
```
_id: ObjectId('69a0c06139df3f9fd2ac989b')
name: "Oussama"
email: "oussama@gmail.com"
__v: 0
```

C. Orders Service (Service des Emprunts)

- **Rôle** : Gère les transactions d'emprunt entre les membres et les livres.
- **Technologie** : Node.js, Express, MongoDB, Axios (pour la communication inter-services).
- **Points clés** : Logique de dates (date d'emprunt et date de retour prévue à +14 jours).
- **Base de données** : Collection orders.

Lien API :

<https://orders-api-atelier5.vercel.app/orders>

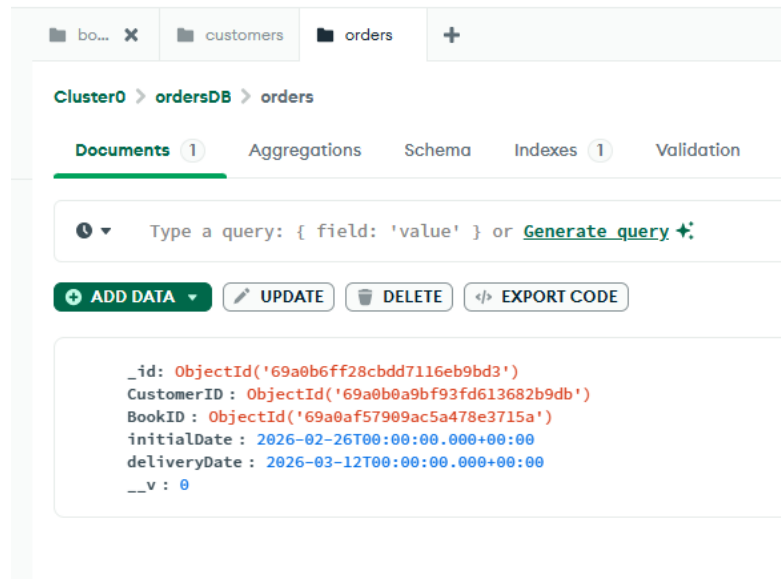


```

[
  {
    "_id": "69a0b6ff28cbdd7116eb9bd3",
    "CustomerID": "69a0b0a9bf93fd613682b9db",
    "BookID": "69a0af57909ac5a478e3715a",
    "initialDate": "2026-02-26T00:00:00.000Z",
    "deliveryDate": "2026-03-12T00:00:00.000Z",
    "__v": 0
  }
]

```

Collection orders :



Pour plus des infromations entrer le GitHub Repository

Pour plus d'informations, consultez le dépôt GitHub.